Class 10 Science — Medium Practice Set

100 Questions | Level: Medium

StudyShelf — studyshelf practice worksheet. Answer key provided at the end.

Q1. Explain why the sky appears blue during the day.

Q2. Explain how a fuse wire protects electrical circuits.

Q3. An electrical appliance operates at 110 V and draws a current of 0.5 A. Find the power consumed.

Q4. Two resistors of 12 Ω and 6 Ω are connected in parallel. Find the equivalent resistance.

Q5. A wave has a frequency of 50 Hz and wavelength 1 m. Find its speed.

Q6. A resistor draws a current of 1.5 A when connected to a 2 V battery. Find its resistance using Ohm's law.

Q7. Three resistors of 7 Ω, 19 Ω and 15 Ω are connected in series. Find the equivalent resistance.

Q8. Two resistors of 2 Ω and 10 Ω are connected in parallel. Find the equivalent resistance.

Q9. Why is vegetative propagation practised for growing sugarcane, roses and grapes?

Q10. A resistor draws a current of 2 A when connected to a 9 V battery. Find its resistance using Ohm's law.

Q11. Explain the reason plants have very few stomata on the upper surface of leaves.

Q12. A wave has a frequency of 50 Hz and wavelength 5 m. Find its speed.

Q13. Three resistors of 20 Ω, 3 Ω and 12 Ω are connected in series. Find the equivalent resistance.

Q14. Three resistors of 12 Ω, 10 Ω and 7 Ω are connected in series. Find the equivalent resistance.

Q15. An electrical appliance operates at 220 V and draws a current of 5 A. Find the power consumed.

Q16. Why is Ohm's law not applicable to all conducting devices?

Q17. A wave has a frequency of 200 Hz and wavelength 1 m. Find its speed.

Q18. An electrical appliance operates at 110 V and draws a current of 2 A. Find the power consumed.

Q19. Three resistors of 3 Ω, 15 Ω and 10 Ω are connected in series. Find the equivalent resistance.

Q20. Three resistors of 11 Ω, 18 Ω and 15 Ω are connected in series. Find the equivalent resistance.

Q21. An electrical appliance operates at 230 V and draws a current of 1 A. Find the power consumed.

Q22. A resistor draws a current of 1.5 A when connected to a 24 V battery. Find its resistance using Ohm's law.

Q23. An electrical appliance operates at 230 V and draws a current of 0.5 A. Find the power consumed.

Q24. A resistor draws a current of 2.5 A when connected to a 24 V battery. Find its resistance using Ohm's law.

Q25. Three resistors of 19 Ω, 18 Ω and 15 Ω are connected in series. Find the equivalent resistance.

Q26. A wave has a frequency of 500 Hz and wavelength 4 m. Find its speed.

Q27. Why do we store silver chloride in dark coloured bottles?

Q28. An electrical appliance operates at 220 V and draws a current of 0.5 A. Find the power consumed.

Q29. A wave has a frequency of 100 Hz and wavelength 5 m. Find its speed.

Q30. Why do desert plants have a special mode of photosynthesis?

Q31. Explain why metals like sodium and potassium are stored in kerosene oil.

Q32. An electrical appliance operates at 110 V and draws a current of 1 A. Find the power consumed.

Q33. Explain why washing soda is used to remove the permanent hardness of water.

Q34. An electrical appliance operates at 220 V and draws a current of 1 A. Find the power consumed.

Q35. A resistor draws a current of 3 A when connected to a 10 V battery. Find its resistance using Ohm's law.

Q36. Why is the process of digestion of starch considered a chemical change?

Q37. Three resistors of 6 Ω, 8 Ω and 5 Ω are connected in series. Find the equivalent resistance.

Q38. Why is the transfer of heat by conduction minimum in gases?

Q39. Three resistors of 16 Ω, 3 Ω and 5 Ω are connected in series. Find the equivalent resistance.

Q40. An electrical appliance operates at 230 V and draws a current of 5 A. Find the power consumed.

Q41. A resistor draws a current of 2 A when connected to a 15 V battery. Find its resistance using Ohm's law.

Q42. Two resistors of 2 Ω and 12 Ω are connected in parallel. Find the equivalent resistance.

Q43. Two resistors of 6 Ω and 2 Ω are connected in parallel. Find the equivalent resistance.

Q44. A wave has a frequency of 500 Hz and wavelength 2 m. Find its speed.

Q45. Why does the colour of copper sulphate solution change when an iron nail is dipped in it?

Q46. Why does the density of water increase up to 4°C and then decrease?

Q47. A resistor draws a current of 1.5 A when connected to a 4 V battery. Find its resistance using Ohm's law.

Q48. Two resistors of 12 Ω and 10 Ω are connected in parallel. Find the equivalent resistance.

Q49. An electrical appliance operates at 230 V and draws a current of 2 A. Find the power consumed.

Q50. Explain why respiration is considered an exothermic reaction.

Q51. A resistor draws a current of 1 A when connected to a 4 V battery. Find its resistance using Ohm's law.

Q52. Three resistors of 10 Ω, 17 Ω and 4 Ω are connected in series. Find the equivalent resistance.

Q53. A resistor draws a current of 2.5 A when connected to a 4 V battery. Find its resistance using Ohm's law.

Q54. Two resistors of 10 Ω and 4 Ω are connected in parallel. Find the equivalent resistance.

Q55. A wave has a frequency of 250 Hz and wavelength 4 m. Find its speed.

Q56. Explain why we cannot see clearly for a few moments when entering a dark room from bright sunlight.

Q57. A resistor draws a current of 3 A when connected to a 4 V battery. Find its resistance using Ohm's law.

Q58. Three resistors of 7 Ω, 7 Ω and 12 Ω are connected in series. Find the equivalent resistance.

Q59. An electrical appliance operates at 110 V and draws a current of 5 A. Find the power consumed.

Q60. A resistor draws a current of 2 A when connected to a 2 V battery. Find its resistance using Ohm's law.

Q61. Two resistors of 2 Ω and 6 Ω are connected in parallel. Find the equivalent resistance.

Q62. Two resistors of 8 Ω and 12 Ω are connected in parallel. Find the equivalent resistance.

Q63. Three resistors of 15 Ω, 6 Ω and 13 Ω are connected in series. Find the equivalent resistance.

Q64. A resistor draws a current of 3 A when connected to a 9 V battery. Find its resistance using Ohm's law.

Q65. Three resistors of 5 Ω, 7 Ω and 19 Ω are connected in series. Find the equivalent resistance.

Q66. Two resistors of 6 Ω and 6 Ω are connected in parallel. Find the equivalent resistance.

Q67. A resistor draws a current of 1 A when connected to a 20 V battery. Find its resistance using Ohm's law.

Q68. A resistor draws a current of 2 A when connected to a 20 V battery. Find its resistance using Ohm's law.

Q69. Two resistors of 8 Ω and 4 Ω are connected in parallel. Find the equivalent resistance.

Q70. Three resistors of 8 Ω, 17 Ω and 14 Ω are connected in series. Find the equivalent resistance.

Q71. A resistor draws a current of 2 A when connected to a 4 V battery. Find its resistance using Ohm's law.

Q72. Three resistors of 6 Ω, 8 Ω and 19 Ω are connected in series. Find the equivalent resistance.

Q73. A resistor draws a current of 0.5 A when connected to a 24 V battery. Find its resistance using Ohm's law.

Q74. Three resistors of 12 Ω, 2 Ω and 5 Ω are connected in series. Find the equivalent resistance.

Q75. Three resistors of 3 Ω, 4 Ω and 5 Ω are connected in series. Find the equivalent resistance.

Q76. Three resistors of 4 Ω, 7 Ω and 10 Ω are connected in series. Find the equivalent resistance.

Q77. Three resistors of 11 Ω, 12 Ω and 15 Ω are connected in series. Find the equivalent resistance.

Q78. Three resistors of 18 Ω, 19 Ω and 12 Ω are connected in series. Find the equivalent resistance.

Q79. A resistor draws a current of 1 A when connected to a 2 V battery. Find its resistance using Ohm's law.

Q80. A resistor draws a current of 0.5 A when connected to a 9 V battery. Find its resistance using Ohm's law.

Q81. A resistor draws a current of 0.5 A when connected to a 4 V battery. Find its resistance using Ohm's law.

Q82. Explain why the nucleus of an atom is positively charged.

Q83. A resistor draws a current of 2 A when connected to a 12 V battery. Find its resistance using Ohm's law.

Q84. A resistor draws a current of 2 A when connected to a 5 V battery. Find its resistance using Ohm's law.

Q85. A resistor draws a current of 2.5 A when connected to a 6 V battery. Find its resistance using Ohm's law.

Q86. A resistor draws a current of 2.5 A when connected to a 20 V battery. Find its resistance using Ohm's law.

Q87. A wave has a frequency of 200 Hz and wavelength 5 m. Find its speed.

Q88. Why does a dilute acid conduct electricity?

Q89. Three resistors of 14 Ω, 4 Ω and 12 Ω are connected in series. Find the equivalent resistance.

Q90. A resistor draws a current of 0.5 A when connected to a 2 V battery. Find its resistance using Ohm's law.

Q91. A wave has a frequency of 250 Hz and wavelength 0.5 m. Find its speed.

Q92. Two resistors of 4 Ω and 2 Ω are connected in parallel. Find the equivalent resistance.

Q93. Two resistors of 4 Ω and 6 Ω are connected in parallel. Find the equivalent resistance.

Q94. Two resistors of 8 Ω and 8 Ω are connected in parallel. Find the equivalent resistance.

Q95. Three resistors of 7 Ω, 14 Ω and 16 Ω are connected in series. Find the equivalent resistance.

Q96. Three resistors of 15 Ω, 10 Ω and 8 Ω are connected in series. Find the equivalent resistance.

Q97. A wave has a frequency of 100 Hz and wavelength 1 m. Find its speed.

Q98. Two resistors of 10 Ω and 6 Ω are connected in parallel. Find the equivalent resistance.

Q99. Two resistors of 6 Ω and 10 Ω are connected in parallel. Find the equivalent resistance.

Q100. Three resistors of 9 Ω, 13 Ω and 16 Ω are connected in series. Find the equivalent resistance.

Answer Key

1. Blue light has a shorter wavelength and gets scattered more by air molecules than other colours (Rayleigh scattering), making the sky appear blue.

2. A fuse wire has a low melting point and high resistance; when excess current flows, it heats up and melts, breaking the circuit and protecting appliances from damage.

3. P = VI = 55.0 W

4. R_total ≈ 4.0 Ω

5. Speed = f × λ = 50 m/s

6. R = V/I = 1.33 Ω

7. R_total = 41 Ω

8. R_total ≈ 1.67 Ω

9. These plants produce true-to-type identical offspring faster, retain desirable characteristics, and can bypass seed formation, making vegetative propagation efficient for them.

10. R = V/I = 4.5 Ω

11. The upper surface receives more direct sunlight, so having fewer stomata there reduces excessive water loss through transpiration.

12. Speed = f × λ = 250 m/s

13. R_total = 35 Ω

14. R_total = 29 Ω

15. P = VI = 1100 W

16. Some devices, like diodes, do not have a constant V/I ratio at all voltages, so their resistance changes and Ohm's law does not strictly hold for them.

17. Speed = f × λ = 200 m/s

18. P = VI = 220 W

19. R_total = 28 Ω

20. R_total = 44 Ω

21. P = VI = 230 W

22. R = V/I = 16.0 Ω

23. P = VI = 115.0 W

24. R = V/I = 9.6 Ω

25. R_total = 52 Ω

26. Speed = f × λ = 2000 m/s

27. Silver chloride decomposes in sunlight into silver and chlorine (a photochemical/decomposition reaction), so dark bottles prevent exposure to light.

28. P = VI = 110.0 W

29. Speed = f × λ = 500 m/s

30. Desert plants like cacti use CAM (Crassulacean Acid Metabolism) photosynthesis, opening stomata at night to reduce water loss in the hot, dry conditions.

31. Sodium and potassium are highly reactive and react vigorously with oxygen and moisture in air, even catching fire, so they are stored under kerosene to prevent contact with air.

32. P = VI = 110 W

33. Washing soda (sodium carbonate) reacts with calcium and magnesium salts present in hard water to form insoluble precipitates, removing them from the water.

34. $P = VI = 220$ W

35. $R = V/I = 3.33\ \Omega$

36. During digestion, enzymes break down starch into simpler sugars, forming new substances with different properties, which makes it a chemical change.

37. $R_{total} = 19\ \Omega$

38. In gases, molecules are far apart and collide less frequently, so heat energy transfer through molecular collision (conduction) is minimal.

39. $R_{total} = 24\ \Omega$

40. $P = VI = 1150$ W

41. $R = V/I = 7.5\ \Omega$

42. $R_{total} \approx 1.71\ \Omega$

43. $R_{total} \approx 1.5\ \Omega$

44. Speed $= f \times \lambda = 1000$ m/s

45. Iron displaces copper from copper sulphate solution (a displacement reaction), forming iron sulphate (light green) and depositing copper, so the blue colour fades.

46. Water shows anomalous expansion — its molecules are most closely packed at 4°C, giving it maximum density, after which further cooling causes ice-like structure formation and expansion.

47. $R = V/I = 2.67\ \Omega$

48. $R_{total} \approx 5.45\ \Omega$

49. $P = VI = 460$ W

50. During respiration, glucose is broken down in the presence of oxygen to release energy along with CO_2 and water, so energy is released, making it exothermic.

51. $R = V/I = 4.0\ \Omega$

52. $R_{total} = 31\ \Omega$

53. $R = V/I = 1.6\ \Omega$

54. $R_{total} \approx 2.86\ \Omega$

55. Speed $= f \times \lambda = 1000$ m/s

56. The pupil takes some time to dilate (widen) to let in more light, and the eye needs time to adapt to the low light condition, causing temporary reduced vision.

57. $R = V/I = 1.33\ \Omega$

58. $R_{total} = 26\ \Omega$

59. $P = VI = 550$ W

60. $R = V/I = 1.0\ \Omega$

61. $R_{total} \approx 1.5\ \Omega$

62. $R_{total} \approx 4.8\ \Omega$

63. $R_{total} = 34\ \Omega$

64. $R = V/I = 3.0\ \Omega$

65. $R_{total} = 31\ \Omega$

66. $R_{total} \approx 3.0\ \Omega$

67. R = V/I = 20.0 Ω

68. R = V/I = 10.0 Ω

69. R_total ≈ 2.67 Ω

70. R_total = 39 Ω

71. R = V/I = 2.0 Ω

72. R_total = 33 Ω

73. R = V/I = 48.0 Ω

74. R_total = 19 Ω

75. R_total = 12 Ω

76. R_total = 21 Ω

77. R_total = 38 Ω

78. R_total = 49 Ω

79. R = V/I = 2.0 Ω

80. R = V/I = 18.0 Ω

81. R = V/I = 8.0 Ω

82. The nucleus contains protons, which carry a positive charge, and neutrons, which are neutral, so overall the nucleus has a net positive charge.

83. R = V/I = 6.0 Ω

84. R = V/I = 2.5 Ω

85. R = V/I = 2.4 Ω

86. R = V/I = 8.0 Ω

87. Speed = f × λ = 1000 m/s

88. A dilute acid ionises in water to produce H+ ions, and the presence of these free ions allows the solution to conduct electricity.

89. R_total = 30 Ω

90. R = V/I = 4.0 Ω

91. Speed = f × λ = 125.0 m/s

92. R_total ≈ 1.33 Ω

93. R_total ≈ 2.4 Ω

94. R_total ≈ 4.0 Ω

95. R_total = 37 Ω

96. R_total = 33 Ω

97. Speed = f × λ = 100 m/s

98. R_total ≈ 3.75 Ω

99. R_total ≈ 3.75 Ω

100. R_total = 38 Ω